

SmogSense: A Spatiotemporal Machine Learning Architecture for Probabilistic PM_{2.5} Forecasting in Sensor-Sparse Regions

Introduction to the Atmospheric and Computational Challenge

The annual degradation of air quality across the Indo-Gangetic Plain represents a profound public health emergency and an intricate atmospheric physics problem. Every winter, vast expanses of northern India and eastern Pakistan, with the metropolitan hub of Lahore at the epicenter, are enveloped in a hazardous layer of particulate matter. This phenomenon is driven by a confluence of localized anthropogenic emissions, massive transboundary agricultural biomass burning, and highly restrictive winter meteorology. The resulting atmospheric

conditions routinely elevate $PM_{2.5}$ to concentrations that vastly exceed safe limits; in one notable episode in November 2024, the Air Quality Index (AQI) in Lahore reached a catastrophic 1900¹. Tackling this crisis aligns directly with the United Nations Sustainable Development Goals, specifically SDG 3 (Good Health and Well-being), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action).

Despite the severity of this annual environmental disaster, the deployment of regulatory-grade, continuous air quality monitoring infrastructure remains perilously sparse relative to the population density and the extreme spatial heterogeneity of the pollution. While the fundamental physics of the smog are understood, predicting its severity 24 to 72 hours in advance at a hyper-local level is computationally difficult. Traditional machine learning architectures rely heavily on dense networks of ground-truth sensors to capture non-linear local interactions. When these models are deployed in sensor-sparse regions, their predictive accuracy degrades exponentially, rendering them useless for actionable public health advisories.

The "SmogSense" architecture addresses this fundamental limitation by proposing a hybrid, spatially-aware machine learning pipeline designed to maintain predictive utility in data-deprived environments. By shifting the predictive burden away from sparse ground sensors and toward ubiquitous, globally available data sources—specifically satellite-derived Aerosol Optical Depth (AOD), numerical atmospheric reanalysis, and active fire hotspot telemetry—the system establishes a robust physical baseline.

The architecture bridges the performance gap between data-rich and data-sparse domains through few-shot transfer learning methodologies. Specifically, the model leverages historical data from densely instrumented cities, such as Delhi, to learn the universal physical priors of the Indo-Gangetic airshed, subsequently fine-tuning these parameters for Lahore using the

limited available local data. Acknowledging the inherent chaos of atmospheric dispersion, the system abandons deterministic point forecasting in favor of probabilistic forecasting via quantile regression. This exhaustive research report delineates the theoretical foundations, data engineering requirements, algorithmic architecture, and deployment strategy for this system, providing a blueprint for a fully automated, scheduled pipeline that publishes localized, bilingual warnings directly to affected populations.

Atmospheric Physics and the Drivers of Winter Smog

To engineer a predictive model that transcends basic autoregressive statistical extrapolation,

the feature space must explicitly encode the physical mechanisms that drive $PM_{2.5}$ accumulation. The winter smog in Lahore is not merely a function of increased local emissions; it is predominantly dictated by specific meteorological phenomena that suppress the atmospheric dispersion of existing pollutants.

Planetary Boundary Layer Dynamics and the Ventilation Coefficient

The planetary boundary layer (PBL) represents the lowest tier of the troposphere, directly influenced by its contact with the planetary surface. During the winter months in the Indo-Gangetic Plain, diminished surface temperatures and reduced solar radiation lead to a drastically shallow boundary layer. When temperature inversions occur—a meteorological anomaly where a layer of warmer air overlies cooler air near the surface—the vertical mixing of pollutants is almost completely restricted.

The capacity of the atmosphere to disperse suspended particulate matter is a function of both vertical mixing height and horizontal transport speed. This relationship is mathematically

formalized by the Ventilation Coefficient (VC), calculated as the product of the Planetary Boundary Layer Height ($PBLH$) and the scalar wind speed (WS)².

$$VC = PBLH \times WS$$

Extensive atmospheric research indicates that during periods of unfavorable winter meteorology in the region, both the planetary boundary layer height and wind speeds drop to seasonal minimums, thereby drastically reducing the ventilation coefficient⁴. The dilution capacity of the planetary boundary layer contributes significantly to the modulation of

$PM_{2.5}$ levels, with low ventilation coefficients resulting in the stagnation of air masses and the dangerous accumulation of particulate matter near the surface⁵. Analysis reveals that the ventilation coefficient can be up to four times lower in winter compared to summer, directly

correlating with periods of extreme toxicity⁴. Consequently, $PBLH$ and vector wind speeds extracted from atmospheric reanalysis datasets serve as high-importance, non-negotiable features for any predictive model operating in this geographical domain.

Transboundary Aerosol Transport and Biomass Burning

While localized urban emissions from vehicular exhaust, industrial facilities, and brick kilns provide a steady baseline of pollution, the extreme spikes in winter $PM_{2.5}$ are heavily influenced by the influx of aerosolized biomass. Following the autumn harvest, the widespread practice of paddy stubble burning in the upwind plains releases massive quantities of black carbon, carbon monoxide, and volatile organic compounds (VOCs)⁷. These VOCs act as critical precursors to secondary organic aerosols, which can constitute up to 60% of the $PM_{2.5}$ mass in the region during peak burning season⁸. These emissions are transported horizontally by regional wind patterns, primarily the north-westerly winds blowing from Indian Punjab toward the southeast. Therefore, capturing the spatial density and temporal intensity of upwind fire hotspots is critical for predicting spikes in $PM_{2.5}$ concentrations 24 to 72 hours in advance. Models must ingest thermal anomalies and calculate temporal lags based on distance and wind velocity to accurately simulate the arrival of these transboundary smoke plumes.

Feasibility and Data Acquisition Strategy

The feasibility of constructing a highly accurate, automated forecasting pipeline relies on the ingestion, harmonization, and preprocessing of disparate, open-access data streams. The proposed architecture synthesizes ground-truth observations, satellite remote sensing, and numerical weather prediction outputs.

Ground-Truth Particulate Matter Data via OpenAQ

The OpenAQ platform serves as the primary repository for ground-level $PM_{2.5}$ measurements, aggregating data from diverse providers into a unified API. A critical requirement for the SmogSense project is assessing the spatial coverage and reliability of this network within Pakistan. Recent environmental initiatives, particularly those spearheaded by The Urban Unit of Pakistan, have significantly expanded the monitoring network by installing 160 low-cost sensors (such as AirGradient Open Air Generation 1 and Clarity devices) across 14 major cities⁹.

Despite this expansion, relying on this data presents acute engineering challenges. Low-cost optical sensors are subject to calibration drifts and environmental interference; for instance, high relative humidity can distort laser scattering mechanisms, artificially inflating $PM_{2.5}$ readings¹¹. Furthermore, power grid instability and internet outages in the region result in frequent missing data intervals.

The OpenAQ API (v3) provides programmatic access but imposes a free-tier rate limit of 60 requests per minute¹². The Python-based data ingestion pipeline must implement robust pagination, exponential backoff for rate-limit handling, and sophisticated missing-value imputation. For short gaps (under 3 hours), spline interpolation is mathematically appropriate; for larger gaps, the pipeline must rely on background imputation from regional numerical

models¹⁴.

Data Provider	Sensor Type	Key Locations	Target Variable	Limitations
OpenAQ (Urban Unit)	AirGradient O-1PST	Lahore (Jail Road, Township)	$PM_{2.5}$, PM_{10} , RH, Temp	Calibration drift, missing temporal sequences due to local outages.
OpenAQ (US EPA)	Regulatory Grade	Lahore US Consulate, Islamabad	$PM_{2.5}$, O_3 , NO_2	Highly reliable but extreme spatial sparsity.

Atmospheric Reanalysis and Background Forecasting

To model the ventilation coefficient and other meteorological drivers, the pipeline integrates data from the Copernicus Climate Data Store (CDS). The ERA5 dataset provides hourly estimates of atmospheric variables, including boundary layer height, temperature, relative humidity, and u/v wind components at a $0.25^\circ \times 0.25^\circ$ spatial resolution¹⁶. Because ERA5 has a latency of approximately five days (released as ERA5T), it is highly suitable for historical model training¹⁶. However, for real-time operational 72-hour forecasting, the pipeline must rely on live numerical weather prediction forecasts (such as NOAA GFS) or the Copernicus Atmosphere Monitoring Service (CAMS) near-real-time global atmospheric composition forecasts¹⁷.

CAMS provides baseline predictions for aerosols and trace gases, offering up to 5-day forecasts¹⁷. The proposed architecture will utilize the CAMS $PM_{2.5}$ regional forecast as a baseline input feature. If the localized machine learning pipeline cannot outperform this physics-based numerical forecast, the added complexity of the AI system is unjustified. The CDS API Python client allows for the programmatic downloading of these datasets in GRIB or NetCDF formats¹⁹. However, the free tier restricts users to specific request quotas, concurrent connections, and monthly transfer limits, necessitating careful geographic subsetting via bounding boxes to avoid exceeding processing units²¹.

Copernicus API Constraint	Limit (Free Tier)	Mitigation Strategy for Automated Pipeline
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Concurrent Connections	4 max	Serialize API requests in GitHub Actions workflow.
Bandwidth per Connection	20 MB/s	Subset spatial bounding box strictly to the IGP.
Token Expiration	10 minutes	Implement dynamic token refresh in Python client.

Satellite Remote Sensing: Aerosol Optical Depth and Fire Anomalies

Aerosol Optical Depth (AOD) measures the extinction of solar radiation by suspended particles. The NASA MODIS instrument provides the MCD19A2 Multi-angle Implementation of Atmospheric Correction (MAIAC) AOD product, which offers high-resolution (1 km) daily

aerosol data²². AOD is mathematically correlated with surface $PM_{2.5}$, making it a powerful spatial feature.

However, a critical algorithmic limitation exists within the AOD retrieval process regarding cloud masking. During extreme smog events, the high optical thickness of the aerosol layer is frequently misclassified by satellite algorithms as cloud cover²⁴. Consequently, the pixels are masked and discarded, resulting in extensive missing data exactly when pollution levels are at their most critical¹⁴. The model pipeline must account for this paradox by either utilizing spatial imputation techniques (such as Partial Convolutional Neural Networks) or by shifting feature importance toward CAMS background data and FIRMS fire counts when AOD retrievals fail¹⁴. To quantify the transboundary biomass burning, the pipeline utilizes the NASA Fire Information for Resource Management System (FIRMS), which provides near real-time active fire telemetry from MODIS and VIIRS sensors²⁶. By defining a geospatial bounding box upwind of Lahore (based on the prevailing north-westerly wind vectors derived from ERA5), the pipeline aggregates daily fire counts and Fire Radiative Power (FRP). These aggregated metrics serve as a highly predictive proxy for the incoming volume of biomass smoke, effectively providing the model with advanced warning of advected pollution²⁷.

Algorithmic Architecture: Sequence Modeling and Gradient Boosting

Predictive modeling of $PM_{2.5}$ in a complex airshed requires an algorithm capable of simultaneously processing tabular meteorological data, static geographic features, and highly non-stationary temporal pollution sequences. A hybrid architecture combining sequential deep learning with gradient-boosted decision trees offers the optimal balance of temporal awareness, non-linear interaction modeling, and computational efficiency.

Feature Engineering and Temporal Extraction

The raw multivariate data streams must first be transformed into a supervised learning format. This is accomplished using a moving window sampling technique²⁸. A lookback window of size L (e.g., the preceding 72 hours of historical $PM_{2.5}$, boundary layer heights, and upwind fire counts) is extracted to predict the target concentration at future time horizons $T + 24, T + 48$, and $T + 72$ ²⁸.

Deep learning approaches, particularly Temporal Convolutional Networks (TCN) or Gated Recurrent Units (GRU), are exceptionally well-suited for extracting latent temporal embeddings from these multivariate time series²⁹. Unlike traditional Recurrent Neural Networks, TCNs utilize causal, dilated convolutions to process long historical sequences efficiently without suffering from vanishing gradients, while GRUs offer a lightweight gating mechanism to maintain memory of distant meteorological events.

However, pure deep learning models frequently struggle to model abrupt, non-linear thresholds in tabular atmospheric data as effectively as tree-based models²⁷.

Gradient-boosted frameworks, such as LightGBM, excel at capturing complex decision boundaries—for example, the non-linear interaction where high upwind fire counts only result in massive local pollution if the wind direction is precisely north-westerly and the planetary boundary layer drops below a critical threshold²⁷.

Therefore, the SmogSense architecture dictates a bifurcated approach. A small-parameter GRU or TCN acts as an encoder, compressing the 72-hour historical time-series into a dense latent vector. This vector is then concatenated with the deterministic meteorological forecasts (from CAMS/GFS) and satellite features for the future timesteps. This fused feature space is fed into a LightGBM regressor to generate the final multi-horizon prediction.

Domain Adaptation and Model-Agnostic Meta-Learning

The core hypothesis driving the research design is that models trained exclusively on well-instrumented cities (the source domain) suffer severe degradation when deployed in sensor-sparse cities (the target domain), but this performance gap can be partially rectified via few-shot fine-tuning. Delhi represents the optimal source domain due to its exceptionally dense network of regulatory-grade monitors and its placement within the identical Indo-Gangetic airshed, subjecting it to identical macroeconomic forcing, winter inversions, and biomass burning²⁷.

The model undergoes extensive pre-training on a massive dataset of Delhi's historical winter smog seasons. However, direct zero-shot transfer to Lahore will yield biased predictions due to micro-climatic disparities, distinct local emission inventories (e.g., the specific placement of the Quaid-e-Azam Industrial Estate relative to monitors), and calibration discrepancies between Delhi's regulatory sensors and Lahore's predominantly low-cost network¹¹.

To overcome this domain shift, the architecture implements Model-Agnostic Meta-Learning (MAML) for few-shot sensor calibration and forecasting. MAML explicitly optimizes the model parameters such that a small number of gradient updates will lead to rapid convergence on a

new task without overfitting¹¹. Mathematically, consider the model f_θ parameterized by θ . For a new target task T_i (representing the limited Lahore dataset), the model parameters are updated via gradient descent on the support set:

$$\theta'_i = \theta - \alpha \nabla_\theta \mathcal{L}_{T_i}(f_\theta)$$

The overarching meta-objective is to find an initial parameter state θ that minimizes the loss across multiple simulated sparse-data tasks. This is expressed as:

$$\min_{\theta} \sum_{T_i \sim p(T)} \mathcal{L}_{T_i}(f_{\theta'_i}) = \sum_{T_i \sim p(T)} \mathcal{L}_{T_i}(f_{\theta - \alpha \nabla_\theta \mathcal{L}_{T_i}(f_\theta)})$$

By calculating the gradient of the gradient (second-order derivatives), MAML learns an internal representation that is broadly applicable to atmospheric physics while remaining highly sensitive to the nuances of a new geographic domain. When the model is finally exposed to Lahore's limited, few-shot data, it adapts rapidly, preserving the universal priors learned from Delhi while tuning to Lahore's specific sensor biases³².

Probabilistic Forecasting and Uncertainty Quantification

Providing a deterministic point forecast (e.g., predicting exactly $185 \mu\text{g}/\text{m}^3$ of $PM_{2.5}$) is scientifically irresponsible in the context of public health advisories, as it conveys a false sense of absolute certainty regarding a chaotic system. To provide actionable intelligence, the model must output a prediction interval—specifically the 10th, 50th (median), and 90th percentiles—communicating both the expected baseline and the worst-case scenario.

Quantile Regression and the Pinball Loss Function

This probabilistic output is achieved through Quantile Regression, utilizing the pinball loss function (also known as quantile loss)³³. Traditional loss functions like Mean Squared Error (MSE) optimize for the conditional mean, while Mean Absolute Error (MAE) optimizes for the conditional median³⁵. Neither provides a confidence bound.

The pinball loss function introduces an asymmetric penalty that forces the optimization algorithm to target specific percentiles of the target distribution. For a desired quantile

$\tau \in (0, 1)$, an actual observation y , and a prediction $\hat{y}$, the pinball loss $\mathcal{L}_\tau$ is defined as:

$$\mathcal{L}_\tau(y, \hat{y}) = \begin{cases} \tau(y - \hat{y}) & \text{if } y \geq \hat{y} \\ (1 - \tau)(\hat{y} - y) & \text{if } y < \hat{y} \end{cases}$$

The physical implication of this piecewise linear function is profound. Consider the 90th percentile forecast ($\tau = 0.90$), which represents the upper bound or "worst-case" scenario.

If the model under-predicts the true pollution level, the penalty multiplier is 0.90 . If it over-predicts, the penalty multiplier is only 0.10 . Under-predictions are penalized nine times more heavily than over-predictions³⁶. This forces the model to push its prediction boundary upward such that 90% of the actual historical observations fall below the predicted line³⁶.

Conversely, for $\tau = 0.10$, over-predictions are heavily penalized, establishing the lower bound of the confidence interval³⁶.

LightGBM natively supports multi-quantile regression. By configuring the objective parameter to 'quantile' and specifying an alpha array (e.g., [0.1, 0.5, 0.9]), the algorithm constructs distinct decision trees tailored to each quantile boundary³⁸. The 50th percentile provides the baseline expectation, while the delta between the 10th and 90th percentiles visualizes the atmospheric uncertainty, allowing hospitals or municipal planners to prepare for potential upper-limit toxicity events³³.

Evaluation Framework and the Sensor-Sparsity Curve

Evaluating a machine learning model designed explicitly for data-scarce environments requires metrics and experimental designs that rigorously test its robustness against data deprivation. Traditional holdout validation is insufficient.

Plotting the Sensor-Sparsity Curve

To validate the central research hypothesis—that few-shot transfer learning mitigates the degradation caused by sensor sparsity—an ablation study must be conducted to plot a "Sensor-Sparsity Curve." This analytical visualization maps the availability of local training data (independent variable) against the forecasting error (dependent variable).

The methodology involves artificially masking the available OpenAQ ground stations in the target domain (Lahore) during the fine-tuning phase. The experiment initializes with zero local

stations, representing a pure zero-shot transfer from Delhi. Subsequently, $1, 2, 5$, and N local stations are incrementally introduced into the MAML support set.

A standard neural network trained from scratch purely on Lahore data will exhibit an exponential decay in error as sensors are added, but will fail catastrophically at $N = 0$ or $N = 1$. The proposed transfer-learning architecture, by contrast, should demonstrate a

flattened curve. It will maintain acceptable error bounds even at $N = 0$ due to the atmospheric physical priors learned from the source domain, with rapid, steady improvements as few-shot data is introduced.

Baselines and Continuous Ranked Probability Score (CRPS)

The model's performance must be contextualized against established baselines. The two primary baselines are:

1. **Persistence Forecast:** The naive assumption that the $PM_{2.5}$ concentration at time $T + 24$ will be identical to time T .
2. **CAMS Numerical Baseline:** The regional deterministic forecast generated by the Copernicus Atmosphere Monitoring Service¹⁷.

Because the architecture produces probabilistic distributions (quantiles) rather than point estimates, Root Mean Square Error (RMSE) is inadequate for evaluation. The Continuous Ranked Probability Score (CRPS) is the premier strictly proper scoring rule for evaluating the accuracy of predictive distributions⁴⁰.

CRPS compares the cumulative distribution function (CDF) of the probabilistic forecast, $F(y)$, with the empirical CDF of the actual observation, x , represented as a Heaviside step function $H(y - x)$ (where $H = 0$ for $y < x$ and $H = 1$ for $y \geq x$)⁴¹. The metric is calculated as the integrated squared difference between these two functions over the entire real line:

$$CRPS(F, x) = \int_{-\infty}^{\infty} (F(y) - H(y - x))^2 dy$$

A lower CRPS indicates a superior probabilistic forecast, penalizing models simultaneously for being inaccurate (the median is far from the truth) and uncalibrated (the intervals are overly wide or confidently wrong)⁴¹. By minimizing CRPS over the validation set, the architecture ensures that the generated prediction intervals are sharp yet reliably encompass the true $PM_{2.5}$ fluctuations⁴³.

Shadow Running and Prospective Validation

Historical backtesting is inherently vulnerable to data leakage and overfitting. Because the smog season in the Indo-Gangetic Plain initiates in late October and peaks through December, deploying the pipeline prospectively allows for genuine out-of-sample validation.

This is achieved via a "shadow run." The model is placed into production, generating daily forecasts for the forthcoming 72 hours. These forecasted quantiles are logged into a secure database. The system then waits 72 hours for the ground-truth observations to materialize on

the OpenAQ network. The predictions are retrospectively scored against reality, providing an honest, unmanipulated assessment of the model's predictive capacity exactly as it functions in a live environment.

Software Engineering Architecture and Automation Pipeline

A highly accurate, probabilistic machine learning model possesses zero public health utility if it remains confined to an academic Jupyter Notebook. The project necessitates a robust, automated software engineering pipeline capable of ingesting live data, executing inference, and disseminating results to the public without daily manual intervention or incurring significant cloud computing overhead.

Utilizing GitHub Actions for CI/CD Compute

Given the financial constraints of an academic research project, leveraging free-tier Continuous Integration and Continuous Deployment (CI/CD) tools is the optimal architectural choice. GitHub Actions provides scheduled workflows via cron triggers, allocating ephemeral virtual environments with sufficient memory and CPU cycles to execute standard inference scripts.

The automated pipeline executes through the following sequence:

1. **Trigger:** A CRON job initiates the workflow daily at 00:00 UTC.
2. **Data Ingestion:** Python scripts utilize the openaq SDK¹⁵ and the cdsapi package¹⁹ to retrieve the latest 72 hours of ground-truth observations and meteorological forecasts. The script dynamically handles pagination and rate-limit delays. FIRMS hotspot telemetry is fetched via bounding box queries¹².
3. **Preprocessing & Imputation:** Missing satellite or sensor data is imputed. Feature engineering scripts dynamically compute the ventilation coefficient and temporal lagged features.
4. **Model Inference:** The pre-trained LightGBM and TCN models are loaded. The inference script generates the 10th, 50th, and 90th percentile predictions for the forthcoming 72 hours³⁹.
5. **Artifact Generation:** The pipeline utilizes Python visualization libraries (matplotlib, seaborn) to render high-quality, shareable graphics (e.g., PNG format) depicting the predictive intervals. These images are specifically optimized in aspect ratio and file size for mobile viewing on WhatsApp. Simultaneously, a templating engine (e.g., Jinja2) injects the numerical forecast data into a static HTML bulletin, dynamically rendering the content in both English and Urdu to ensure maximum local accessibility.
6. **Deployment:** The generated HTML files and images are automatically committed and pushed to the gh-pages branch of the repository, instantaneously updating a public-facing static website hosted on GitHub Pages at zero cost.

Dissemination and Expert Critique

The automated creation of a shareable image tailored for WhatsApp is a critical socio-technical feature, as WhatsApp serves as the primary vector for rapid information dissemination in South Asia. To validate the utility and scientific integrity of the generated bulletin, engaging an

environmental scientist or a local climate journalist to critique the output is an essential component of the MVP iteration. Their qualitative feedback regarding the clarity of the uncertainty intervals—ensuring the lay public comprehends the physical meaning of a "90th percentile worst-case scenario"—and the actionable nature of the warnings will inform vital UI/UX improvements prior to peak smog season.

Strategic Scaling: Analyzing the "Whole Pakistan"

Feasibility

A critical consideration in the conceptualization of this system is whether to expand the spatial domain from the city of Lahore to the entirety of Pakistan. While scaling the model nationally is theoretically appealing from an impact perspective, an exhaustive analysis of the data landscape and atmospheric physics suggests severe, potentially insurmountable limitations for a budget-constrained academic project.

Spatial Discrepancies in Sensor Coverage

The primary obstacle to a national model is the extreme spatial heterogeneity of the ground-truth training data. An analysis of the OpenAQ database reveals that the vast majority of continuous monitoring stations in Pakistan are heavily clustered within the Punjab province. This is largely attributed to the localized deployment of sensors by The Urban Unit⁹. While the capital, Islamabad, and the southern megacity of Karachi (Sindh) possess moderate coverage, vast geographical expanses including Balochistan, Khyber Pakhtunkhwa, and rural Sindh possess near-zero continuous monitoring infrastructure¹⁰.

Attempting to train a uniform national machine learning model over areas with zero ground truth to validate against introduces unquantifiable predictive errors. Without local fine-tuning data, the model would devolve into merely downscaling CAMS or MERRA-2 reanalysis data, rendering the predictions highly unreliable in varied topographies.

Topographical and Climatological Heterogeneity

Pakistan encompasses a radical diversity of micro-climates. The meteorological parameters governing air quality in the coastal, humid, sea-breeze-dominated environment of Karachi are fundamentally distinct from the landlocked, winter-inversion-dominated atmospheric dynamics of the Indo-Gangetic Plain in Lahore, or the arid, high-altitude terrain of Quetta¹. A single monolithic LightGBM/TCN model trained primarily on Punjab data will suffer from

extreme covariate shift when attempting to predict $PM_{2.5}$ in coastal or mountainous regions. The relationships between wind speed, temperature, boundary layer height, and particulate accumulation are not globally stationary; they are highly dependent on local topography and specific regional emission inventories⁵.

Recommendation for Project Scope

Given the constraints of a final-year academic project, expanding to the entirety of Pakistan risks diluting the methodological rigor of the research. The core scientific contributions—the sensor-sparsity curve, few-shot MAML transfer learning, and quantile regression for uncertainty quantification—are best demonstrated in a well-defined, geographically

constrained problem space.

The optimal strategy is a highly focused approach. The Minimum Viable Product (MVP) should strictly target the Lahore metropolitan area, utilizing Delhi as the pre-training source domain. If time and computational resources permit, the scope can be marginally expanded to encompass the broader Punjab province, as the topography remains relatively flat and the primary forcing mechanisms (agricultural stubble burning and winter atmospheric inversions) remain consistent across the region². Expanding to the whole of Pakistan should be reserved for future post-graduate research, requiring a federated learning approach or the deployment of discrete, region-specific expert models.

Target Publication Venues

The methodology, empirical findings, and software artifacts generated by the SmogSense project present significant contributions to the intersection of applied machine learning, atmospheric science, and computational sustainability. Appropriate venues for publication include:

- **NeurIPS Tackling Climate Change with Machine Learning (CCAI):** The premier workshop for demonstrating how novel ML architectures (such as few-shot MAML and quantile regression) address specific, urgent climate and environmental crises. The empirical validation of the sensor-sparsity curve is highly relevant to this audience.
- **ACM COMPASS (Computing and Sustainable Societies):** This venue focuses on computational research directed toward societal challenges in the developing world. The low-cost deployment pipeline, the bilingual public bulletin, and the focus on data democratization in the Global South make this an ideal fit.
- **IEEE IGARSS or Remote Sensing of Environment (RSE):** If the research pivots heavily toward the specific methodology of imputing MODIS MCD19A2 AOD data during dense smog events and fusing it with FIRMS telemetry, these remote sensing journals serve as highly appropriate targets.

Synthesis and Architectural Outlook

The SmogSense architecture represents a sophisticated, mathematically rigorous approach to democratizing air quality forecasting in sensor-sparse environments. By fusing globally available satellite telemetry and meteorological reanalysis streams with advanced machine learning techniques, the system elegantly circumvents the traditional requirement for prohibitively expensive, dense sensor networks.

The utilization of Model-Agnostic Meta-Learning enables the transfer of complex atmospheric dynamics learned from data-rich neighbors like Delhi, while few-shot fine-tuning anchors the model to the micro-climatic realities of Lahore. Furthermore, the explicit emphasis on probabilistic forecasting via the pinball loss function ensures that the public and policymakers are provided with an honest assessment of atmospheric uncertainty, moving the field beyond flawed deterministic point estimates. Deployed entirely on zero-cost infrastructure through automated GitHub Actions, the generation of accessible, bilingual bulletins demonstrates a profound understanding of how to bridge the gap between high-level computational research and tangible societal impact. By constraining the initial geographic scope to the Indo-Gangetic

Plain, the project guarantees scientific rigor, ensuring that the resulting sensor-sparsity analysis provides a definitive contribution to the field of computational sustainability and environmental health.

Works cited

1. The Power of a Large Air Sensor Network in Pakistan | by OpenAQ, <https://openaq.medium.com/the-power-of-a-large-air-sensor-network-in-pakistan-7a919343a32f>
2. Submicron aerosol composition in the world's most polluted megacity, <https://acp.copernicus.org/articles/19/6843/2019/acp-19-6843-2019.html>
3. Reactive chlorine-, sulfur-, and nitrogen-containing volatile organic, <https://acp.copernicus.org/articles/24/13129/2024/>
4. Submicron aerosol composition in the world's most polluted megacity, <https://acp.copernicus.org/preprints/acp-2018-1066/acp-2018-1066-manuscript-version2.pdf>
5. Statistical analysis for the effect of meteorological factors and, https://www.researchgate.net/publication/395468530_Statistical_analysis_for_the_effect_of_meteorological_factors_and_pollutant_oxides_on_particulate_matter_using_multiple_linear_regression_model
6. IASTA BULLETIN, https://iasta.org.in/images/pdf/Proceedings_IASTA_2012.pdf
7. Time series of post-monsoon CRB smoke emissions impacts over, https://www.researchgate.net/figure/Time-series-of-post-monsoon-CRB-smoke-emissions-impacts-over-Delhi-under-different-CRB_fig3_343025344
8. Discovery of reactive chlorine, sulphur and nitrogen - EGU sphere, <https://egusphere.copernicus.org/preprints/2024/egusphere-2024-500/egusphere-2024-500.pdf>
9. ARC, Lahore - OpenAQ Explorer, <https://explore.openaq.org/locations/6125629>
10. The Urban Unit of Pakistan Installs 160 Air Quality Monitors, <https://climate.uchicago.edu/news/the-urban-unit-of-pakistan-installs-160-air-quality-monitors-first-effort-of-its-kind-in-the-country/>
11. Few-shot calibration of low-cost air pollution (PM2.5) sensors using, <https://arxiv.org/pdf/2108.00640>
12. Access for All: Protecting OpenAQ's Platform and Community, <https://openaq.medium.com/openaq-envisions-a-world-where-everyone-breathes-healthy-air-enabled-by-universal-access-to-data-8f580820cf5d>
13. Rate limits | OpenAQ Docs, <https://docs.openaq.org/using-the-api/rate-limits>
14. Remote Sensing of Aerosols From Space: Retrieval of Properties, https://www.researchgate.net/publication/305722946_Remote_Sensing_of_Aerosols_From_Space_Retrieval_of_Properties_and_Applications
15. About - OpenAQ Python SDK, <https://python.openaq.org/introduction/about/>
16. ERA5 hourly data on single levels from 1940 to present, <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview>
17. Global CAMS data - View Source,

- <https://confluence.ecmwf.int/plugins/viewsource/viewpagesrc.action?pageId=56659592>
18. ECMWF's Code for Earth: Driving solutions in atmospheric monitoring, <https://atmosphere.copernicus.eu/ecmwfs-code-earth-driving-solutions-atmospheric-monitoring>
 19. Downloading ERA-5 Data - Reto Stauffer, <https://retostauffer.org/code/Download-ERA5/>
 20. Complete ERA5 global atmospheric reanalysis - Climate Data Store, https://cds.climate.copernicus.eu/datasets/reanalysis-era5-complete?tab=d_download
 21. Quotas and Limitations - Documentation - Copernicus, <https://documentation.dataspace.copernicus.eu/Quotas.html>
 22. NASA Datasets Available in Cloud Optimized GeoTIFFs, <https://www.earthdata.nasa.gov/about/nasa-datasets-available-cloud-optimized-geotiffs>
 23. Microphysical evolution and column loading drive nonlinear, <https://acp.copernicus.org/articles/26/9149/2026/acp-26-9149-2026.pdf>
 24. Microphysical evolution and column loading drive nonlinear ... - ACP, <https://acp.copernicus.org/articles/26/9149/2026/>
 25. Estimation of Ground PM_{2.5} Concentrations in Pakistan Using, <https://www.mdpi.com/2072-4292/14/7/1735>
 26. Wildfire Early Warning and Rapid Response - NASA Lifelines, <https://nasalifelines.org/hfs-data-series/wildfire-early-warning-and-rapid-response/>
 27. Deep transfer learning and attention based PM_{2.5} forecasting in Delhi, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12394441/>
 28. PM_{2.5} Concentration Prediction in the Cities of China Using Multi, <https://www.mdpi.com/2071-1050/17/19/8891>
 29. Predicting Spatiotemporal Mobile Sensing-Based PM_{2.5} ... - arXiv, <https://arxiv.org/pdf/2609.04693>
 30. A Guide to Adaptation Action Plans to Control Pollution and Smog, <https://www.mdpi.com/2076-3417/12/10/5102>
 31. H-CLAS: A Hybrid Continual Learning Framework for Adaptive Fault, <https://www.mdpi.com/2624-831X/7/1/12>
 32. A CNN-LSTM Model Based on a Meta-Learning Algorithm to Predict, <https://www.mdpi.com/2073-4441/14/15/2377>
 33. Uncertainty in Forecasting: Pinball loss - Kaggle, <https://www.kaggle.com/code/aliakbaryaghoubi/uncertainty-in-forecasting-pinball-loss>
 34. Pinball Loss Function Definition - Lokad, <https://www.lokad.com/pinball-loss-function-definition/>
 35. A Beginner's Deep Dive into Pinball Loss and Quantile Regression, <https://medium.com/@mahendrakumar24325/from-average-predictions-to-real-world-decisions-a-beginners-deep-dive-into-pinball-loss-and-dae66da6e2fc>
 36. Part 02 — The Math Behind the Magic: Understanding the Pinball Loss,

- <https://medium.com/@anik.k.jha/the-math-behind-the-magic-understanding-the-pinball-loss-bb0190d47a9e>
37. An Introduction to Quantile Loss, a.k.a. the Pinball Loss,
<https://towardsdatascience.com/an-introduction-to-quantile-loss-a-k-a-the-pinball-loss-33cccac378a9/>
 38. Quantile Regression in Python with LightGBM: Predicting ... - Medium,
https://medium.com/@suraj_bansal/quantile-regression-in-python-with-lightgbm-predicting-percentiles-part-1-460e6756a053
 39. Prediction intervals explained: A LightGBM tutorial - IBM Developer,
<https://developer.ibm.com/articles/prediction-intervals-explained-a-lightgbm-tutorial/>
 40. properscoring/properscoring/_crps.py at master - GitHub,
https://github.com/TheClimateCorporation/properscoring/blob/master/properscoring/_crps.py
 41. Cannot reproduce Continuous Ranked Probability Score (CRPS),
<https://stats.stackexchange.com/questions/657127/cannot-reproduce-continuous-ranked-probability-score-crps-from-python-package>
 42. Essential Guide to Continuous Ranked Probability Score (CRPS) for,
<https://towardsdatascience.com/essential-guide-to-continuous-ranked-probability-score-crps-for-forecasting-ac0a55dcb30d/>
 43. properscoring/properscoring: Proper scoring rules in Python - GitHub,
<https://github.com/properscoring/properscoring>
 44. The Urban Unit: Real-time Air Quality Sensor Network,
<https://aqicn.org/network/urbanunit/>
 45. situational analysis of air quality in lahore | wwfi,
https://wwfasia.awsassets.panda.org/downloads/situational-analysis-of-air-quality-of-lahore_1_1.pdf